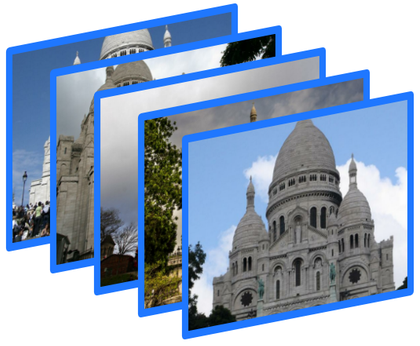
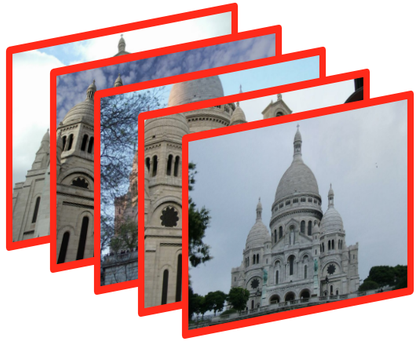
CAPRI

Are two projective reconstructions of the same points consistent? Ask only frame-free numbers.

1 Two view subsets



V_A five photos



V_B five other photos

2 Two projective reconstructions

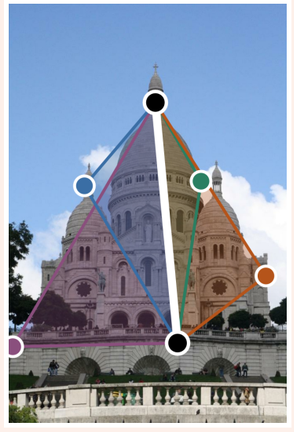


Structure from motion (COLMAP) on all 10 photos gives the cameras. Then each subset triangulates the 565 shared points on its own: two measurements A and B of the same points.

Each is known only up to a frame and a scale per point:

$$p_k \mapsto \lambda_k H\, p_k \quad (H \text{ any } 4\times4)$$

3 Pencil cross-ratios



$$\mathrm{CR}_{ij} = \frac{[ijce]\ [ijdf]}{[ijcf]\ [ijde]}$$

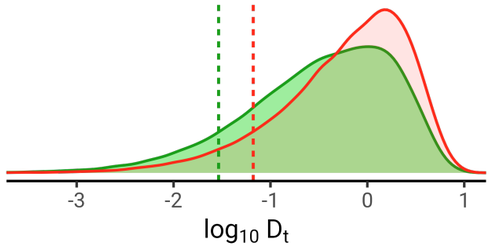
Four planes through the line p_ip_j (white) have a cross-ratio, built from brackets $[ijkl] = \det(p_i, p_j, p_k, p_l)$. Every frame and every point scale cancels.

n points: $3n - 15$ independent ones

4 Consistency test

$$S = Q_{0.1}(|\log|\mathrm{CR}_t(A)| - \log|\mathrm{CR}_t(B)||)$$

low quantile of the differences D_t over all pencils



consistent S = 0.029

170 points jittered by 8 px S = 0.066

Dashed lines: S over 49,991 pencils. The jitter is on the V_B detections of 30% of the points.

$S \le c$	$S > c$
consistent	inconsistent

No frame, no cameras, no noise level: one threshold serves every noise scale.

Illustration on 10 public photos of the Sacré-Cœur (hloc sample data; YFCC100M / IMC, Flickr Creative Commons): 1,516 model points, 565 seen twice in each subset.
It is not an evaluation. The evaluation of CAPRI is on synthetic data and pre-registered; there a known-noise GLRT and the Sampson error match or beat the test.
Photos tilted for display; camera distances in panel 2 are shortened for display.